

Mid-term Exam (Graph Neural Networks - Fall 2026)

Full Name:

Student ID:

Note: Students should write in English for evaluation.

Practice Set 1

10. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

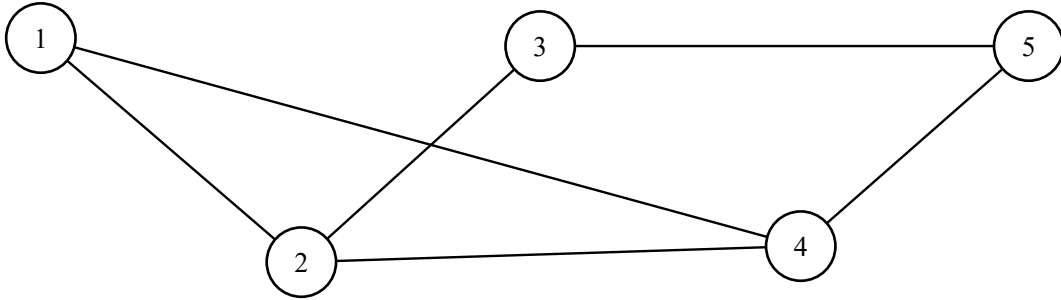


Figure. Graph G

- a) Explain the key assumption behind relational classification, and state one limitation.
- b) Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- c) Calculate $P(Y_v)$ after the 2nd iteration, using the previous iteration's values only.

11. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$ and the t_l nodes are drawn once per layer from $q(u) \propto \|\tilde{A}(:, u)\|^2$, with $\text{ReLU}(x) = \max(0, x)$.

- How does layer-wise sampling differ from the node-wise sampling of GraphSAGE?
- Calculate the importance distribution $q(u)$ over the five nodes.
- Suppose the sampled set is $\{C, D\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

12. (ASGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} -2 \\ -1 \\ 3 \\ 2 \\ 4 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $\tilde{A} = D^{-1}A$, $N(v_i) = \sum_j \tilde{A}_{ij}$, $p(u_j | v_i) = \tilde{A}_{ij} / N(v_i)$, $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- What does the proposal distribution depend on in ASGCN that it does not in FastGCN?
- Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = E$.
- Calculate $E_{p(u_j | E)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(E)$.

13. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

APPNP propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.2$. No nonlinearity is applied between steps.

- Explain the teleport mechanism and why it prevents over-smoothing.
- Calculate $\tilde{A}$ and $Z^{(1)}$.
- Calculate $Z^{(2)}$.

14. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 4 \\ -2 \\ 2 \\ 2 \end{bmatrix}$$

GPR-GNN separates feature extraction from topological propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, $K = 2$, $\gamma_0 = 0.7$, $\gamma_1 = 0.3$, $\gamma_2 = -0.2$.

- What can GPR-GNN do that APPNP cannot, and why do negative γ_k matter?
- Calculate $H^{(1)}$ and $H^{(2)}$.
- Calculate Z .

15. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -3 \\ 3 \\ -1 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \parallel_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- State Theorem 1 of MixHop about the two-hop Delta Operator.
- Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- Calculate $H^{(1)}$ and the Delta Operator $\sigma(\hat{A}H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

16. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 2 & -2 \\ 4 & -1 \\ 4 & 3 \\ 3 & 2 \end{bmatrix}$$

PairNorm normalizes them with a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_i x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_i \|x_i^c\|_2^2} \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- Explain node-wise and feature-wise over-smoothing, and what PairNorm keeps constant.
- Calculate the centered features.
- Calculate the scaling denominator and the final x_i .

17. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -3 \\ 3 \\ 2 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every epoch:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- Which two problems does DropEdge target, and how does it differ from Dropout?
- Calculate the expected number of surviving edges, and $H^{(1)}$ on the full graph.
- Suppose the edges (B, C) and (A, C) are dropped. Recalculate $H^{(1)}$.

1. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

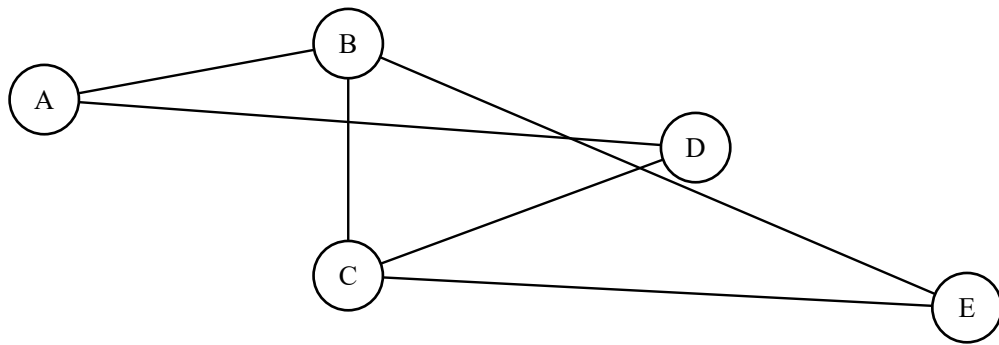


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where $\alpha = 1/p$ if $d(A, x) = 0$, $\alpha = 1$ if $d(A, x) = 1$, and $\alpha = 1/q$ if $d(A, x) = 2$.

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate π_{Bx} .
- Calculate the normalized transition probabilities.

2. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = 1, u'_B = 0.5, u'_C = -1, u'_D = -0.5.$$

The second-order proximity is $p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x)$. Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|C)$, $p_2(B|C)$, $p_2(C|C)$ and $p_2(D|C)$.

3. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 3 \\ -2 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how information is propagated in a GCN layer.
- b) Calculate $\tilde{A}$.
- c) Calculate the output of the GCN model at layer $k = 1$.

4. (GraphSAGE) Consider an undirected graph G of five nodes given in the following figure. Each node has initial features that are the numbers standing next to it. According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

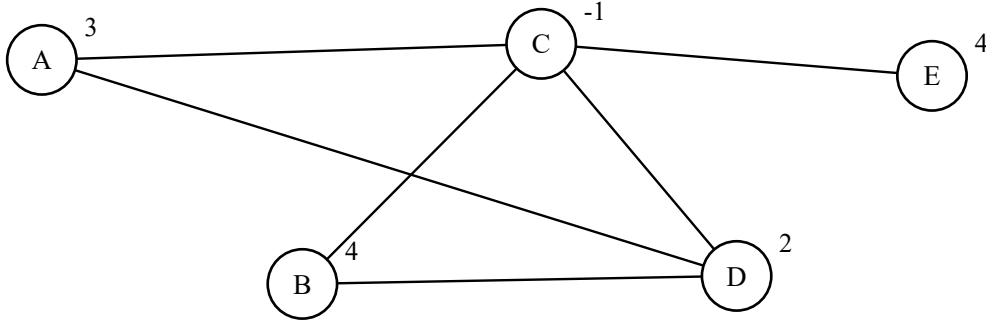


Figure. Graph G

- What is the main idea behind GraphSAGE, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

5. (ClusterGCN) Consider an undirected graph G of six nodes given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

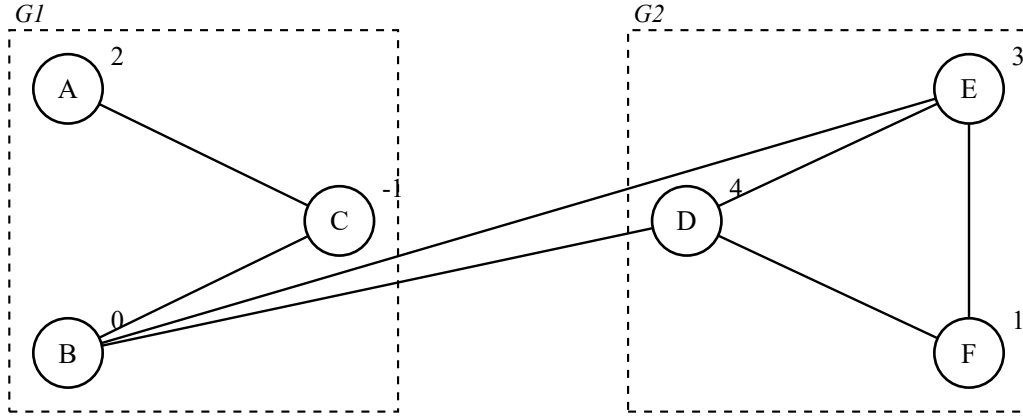


Figure. Graph G

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

6. (GraphSAINT) Consider an undirected graph G of six nodes given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

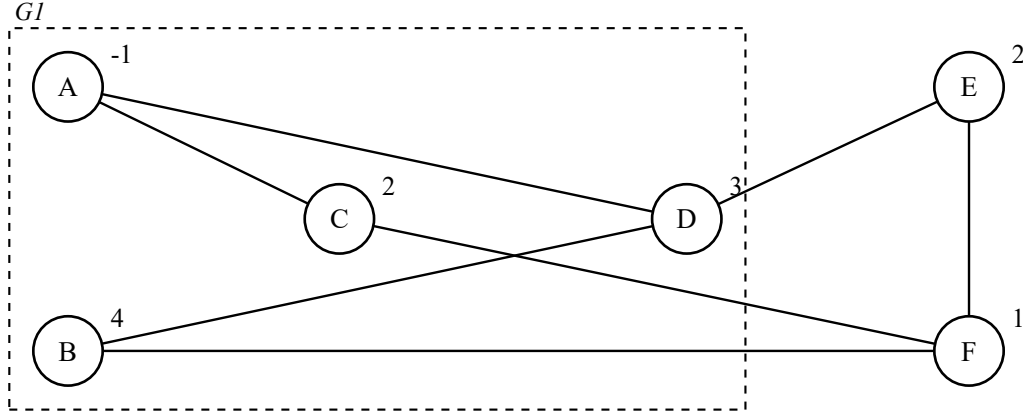


Figure. Graph G

- How does GraphSAINT's sampling differ from node-wise or layer-wise sampling?
- Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -2 \\ 2 \\ -3 \\ 2 \\ 1 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} H^{(0)}), \sigma(\tilde{A} H^{(1)}), \dots, \sigma(\tilde{A} H^{(k-1)}))$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem.
- Calculate $\tilde{A}$ and $H^{(1)}$.
- Calculate the output representations at layer $k = 2$.

8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -3 \\ 0 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $\tilde{A} = D^{-1}A$, I_n is the identity matrix, $\alpha = 0.2$, $\beta = 0.5$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- What are the two major components introduced in GCNII to overcome the depth limitation?
- Calculate $\tilde{A}$.
- Calculate the output representations at layer $k = 1$.

9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 3 \\ 3 \\ 1 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection at layer (k) is:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain why a residual connection allows a GCN to be made deeper.
- Calculate the output representations at layer $k = 1$ and $k = 2$.
- Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.

Practice Set 2

10. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

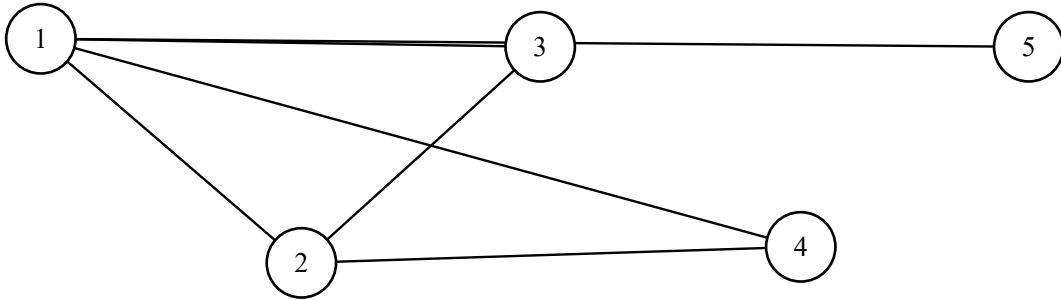


Figure. Graph G

- Explain the key assumption behind relational classification, and state one limitation.
- Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- Calculate $P(Y_v)$ after the 2nd iteration, using the previous iteration's values only.

11. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} -2 \\ -2 \\ 3 \\ 1 \\ -1 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$ and the t_l nodes are drawn once per layer from $q(u) \propto \|\tilde{A}(:, u)\|^2$, with $\text{ReLU}(x) = \max(0, x)$.

- How does layer-wise sampling differ from the node-wise sampling of GraphSAGE?
- Calculate the importance distribution $q(u)$ over the five nodes.
- Suppose the sampled set is $\{D, E\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

12. (ASGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 4 \\ -1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $\tilde{A} = D^{-1}A$, $N(v_i) = \sum_j \tilde{A}_{ij}$, $p(u_j | v_i) = \tilde{A}_{ij} / N(v_i)$, $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- What does the proposal distribution depend on in ASGCN that it does not in FastGCN?
- Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = C$.
- Calculate $E_{p(u_j | C)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(C)$.

13. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

APPNP propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.5$. No nonlinearity is applied between steps.

- a) Explain the teleport mechanism and why it prevents over-smoothing.
- b) Calculate $\tilde{A}$ and $Z^{(1)}$.
- c) Calculate $Z^{(2)}$.

14. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ -2 \end{bmatrix}$$

GPR-GNN separates feature extraction from topological propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, $K = 2$, $\gamma_0 = 0.7$, $\gamma_1 = 0.3$, $\gamma_2 = -0.2$.

- What can GPR-GNN do that APPNP cannot, and why do negative γ_k matter?
- Calculate $H^{(1)}$ and $H^{(2)}$.
- Calculate Z .

15. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ -1 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \parallel_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- State Theorem 1 of MixHop about the two-hop Delta Operator.
- Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- Calculate $H^{(1)}$ and the Delta Operator $\sigma(\hat{A} H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

16. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 0 & 4 \\ 1 & 3 \\ 1 & -2 \\ -2 & 1 \end{bmatrix}$$

PairNorm normalizes them with a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_i x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_i \|x_i^c\|_2^2} \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- Explain node-wise and feature-wise over-smoothing, and what PairNorm keeps constant.
- Calculate the centered features.
- Calculate the scaling denominator and the final x_i .

17. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 0 \\ -2 \\ 2 \\ 2 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every epoch:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- Which two problems does DropEdge target, and how does it differ from Dropout?
- Calculate the expected number of surviving edges, and $H^{(1)}$ on the full graph.
- Suppose the edges (C, D) and (A, C) are dropped. Recalculate $H^{(1)}$.

1. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

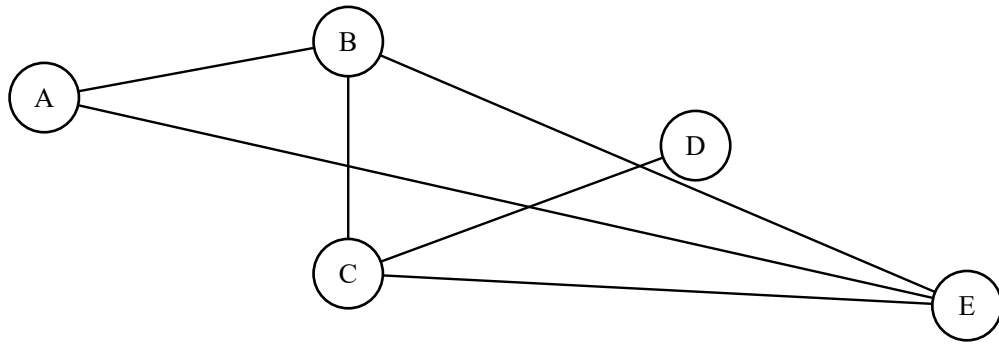


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where $\alpha = 1/p$ if $d(A, x) = 0$, $\alpha = 1$ if $d(A, x) = 1$, and $\alpha = 1/q$ if $d(A, x) = 2$.

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate π_{Bx} .
- Calculate the normalized transition probabilities.

2. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = -0.5, u'_B = 1, u'_C = 0.5, u'_D = -1.$$

The second-order proximity is $p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x)$. Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|C)$, $p_2(B|C)$, $p_2(C|C)$ and $p_2(D|C)$.

3. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ -2 \\ 2 \\ -2 \\ 4 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how information is propagated in a GCN layer.
- Calculate $\tilde{A}$.
- Calculate the output of the GCN model at layer $k = 1$.

4. (GraphSAGE) Consider an undirected graph G of five nodes given in the following figure. Each node has initial features that are the numbers standing next to it. According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

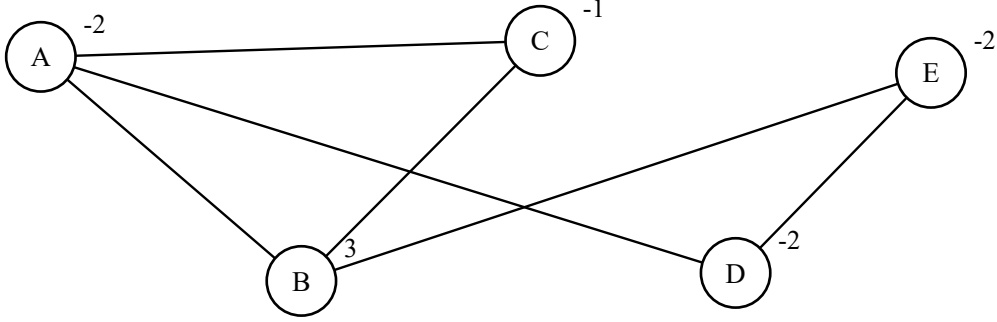


Figure. Graph G

- What is the main idea behind GraphSAGE, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

5. (ClusterGCN) Consider an undirected graph G of six nodes given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

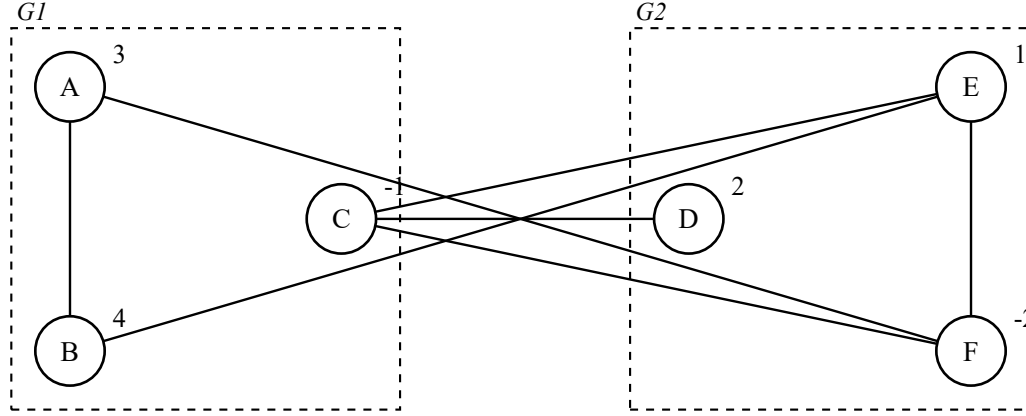


Figure. Graph G

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

6. (GraphSAINT) Consider an undirected graph G of six nodes given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

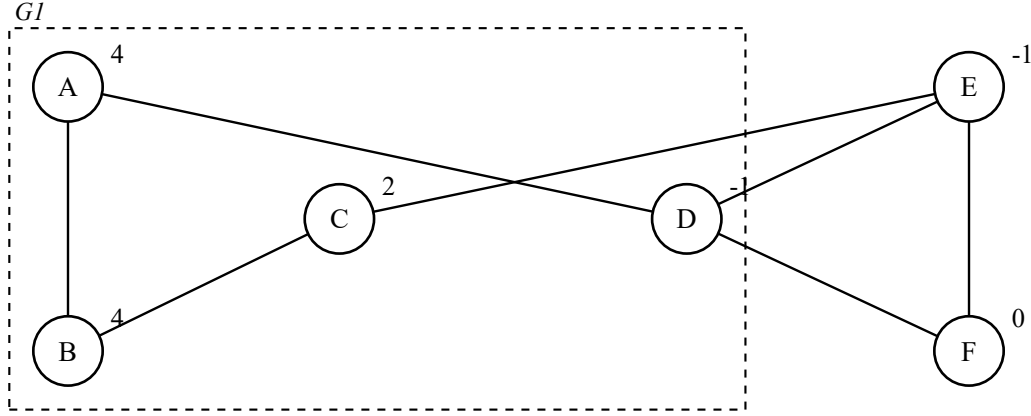


Figure. Graph G

- How does GraphSAINT's sampling differ from node-wise or layer-wise sampling?
- Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ 0 \\ -3 \\ -1 \\ -1 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} H^{(0)}), \sigma(\tilde{A} H^{(1)}), \dots, \sigma(\tilde{A} H^{(k-1)}))$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem.
- Calculate $\tilde{A}$ and $H^{(1)}$.
- Calculate the output representations at layer $k = 2$.

8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 0 \\ -3 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $\tilde{A} = D^{-1}A$, I_n is the identity matrix, $\alpha = 0.2$, $\beta = 0.4$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- What are the two major components introduced in GCNII to overcome the depth limitation?
- Calculate $\tilde{A}$.
- Calculate the output representations at layer $k = 1$.

9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -2 \\ 0 \\ 3 \\ 1 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection at layer (k) is:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain why a residual connection allows a GCN to be made deeper.
- Calculate the output representations at layer $k = 1$ and $k = 2$.
- Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.

Practice Set 3

10. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

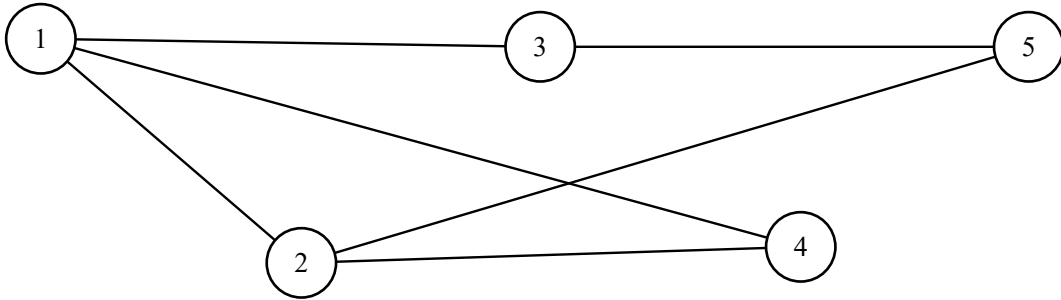


Figure. Graph G

- Explain the key assumption behind relational classification, and state one limitation.
- Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- Calculate $P(Y_v)$ after the 2nd iteration, using the previous iteration's values only.

11. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 1 \\ 0 \\ -2 \\ -1 \\ 2 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$ and the t_l nodes are drawn once per layer from $q(u) \propto \|\tilde{A}(:, u)\|^2$, with $\text{ReLU}(x) = \max(0, x)$.

- How does layer-wise sampling differ from the node-wise sampling of GraphSAGE?
- Calculate the importance distribution $q(u)$ over the five nodes.
- Suppose the sampled set is $\{B, C\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

12. (ASGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} -1 \\ -1 \\ 4 \\ 1 \\ -2 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $\tilde{A} = D^{-1}A$, $N(v_i) = \sum_j \tilde{A}_{ij}$, $p(u_j | v_i) = \tilde{A}_{ij} / N(v_i)$, $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- What does the proposal distribution depend on in ASGCN that it does not in FastGCN?
- Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = C$.
- Calculate $E_{p(u_j | C)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(C)$.

13. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

APPNP propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.5$. No nonlinearity is applied between steps.

- Explain the teleport mechanism and why it prevents over-smoothing.
- Calculate $\tilde{A}$ and $Z^{(1)}$.
- Calculate $Z^{(2)}$.

14. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 4 \\ -2 \\ -1 \\ 4 \end{bmatrix}$$

GPR-GNN separates feature extraction from topological propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, $K = 2$, $\gamma_0 = 0.5$, $\gamma_1 = 0.2$, $\gamma_2 = -0.2$.

- What can GPR-GNN do that APPNP cannot, and why do negative γ_k matter?
- Calculate $H^{(1)}$ and $H^{(2)}$.
- Calculate Z .

15. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ -2 \\ 1 \\ -1 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \parallel_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- State Theorem 1 of MixHop about the two-hop Delta Operator.
- Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- Calculate $H^{(1)}$ and the Delta Operator $\sigma(\hat{A}H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

16. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 4 & 0 \\ 1 & 3 \\ 1 & -1 \\ -1 & 4 \end{bmatrix}$$

PairNorm normalizes them with a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_i x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_i \|x_i^c\|_2^2} \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- a) Explain node-wise and feature-wise over-smoothing, and what PairNorm keeps constant.
- b) Calculate the centered features.
- c) Calculate the scaling denominator and the final x_i .

17. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 0 \\ -3 \\ 4 \\ 3 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every epoch:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- Which two problems does DropEdge target, and how does it differ from Dropout?
- Calculate the expected number of surviving edges, and $H^{(1)}$ on the full graph.
- Suppose the edges (A, C) and (B, D) are dropped. Recalculate $H^{(1)}$.

1. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

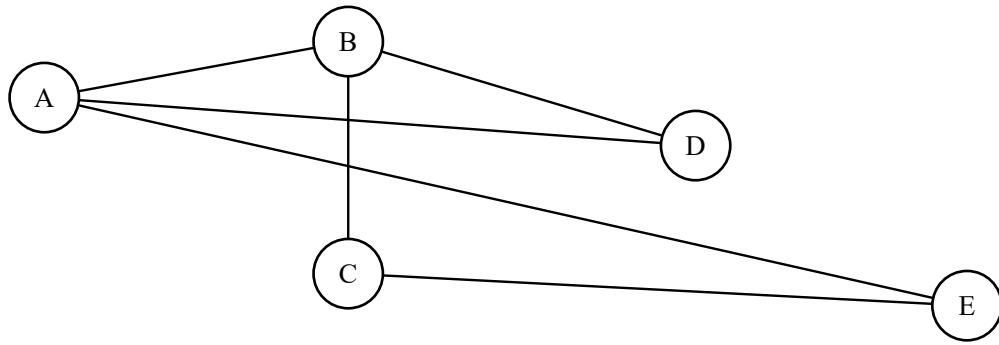


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where $\alpha = 1/p$ if $d(A, x) = 0$, $\alpha = 1$ if $d(A, x) = 1$, and $\alpha = 1/q$ if $d(A, x) = 2$.

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate π_{Bx} .
- Calculate the normalized transition probabilities.

2. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = -0.5, u'_B = 0.5, u'_C = 1, u'_D = -1.$$

The second-order proximity is $p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x)$. Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|B)$, $p_2(B|B)$, $p_2(C|B)$ and $p_2(D|B)$.

3. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -2 \\ -1 \\ 4 \\ 4 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how information is propagated in a GCN layer.
- Calculate $\tilde{A}$.
- Calculate the output of the GCN model at layer $k = 1$.

4. (GraphSAGE) Consider an undirected graph G of five nodes given in the following figure. Each node has initial features that are the numbers standing next to it. According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

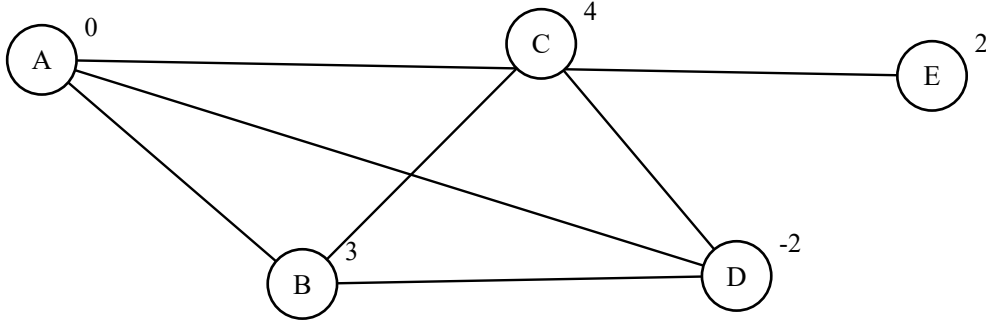


Figure. Graph G

- What is the main idea behind GraphSAGE, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

5. (ClusterGCN) Consider an undirected graph G of six nodes given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

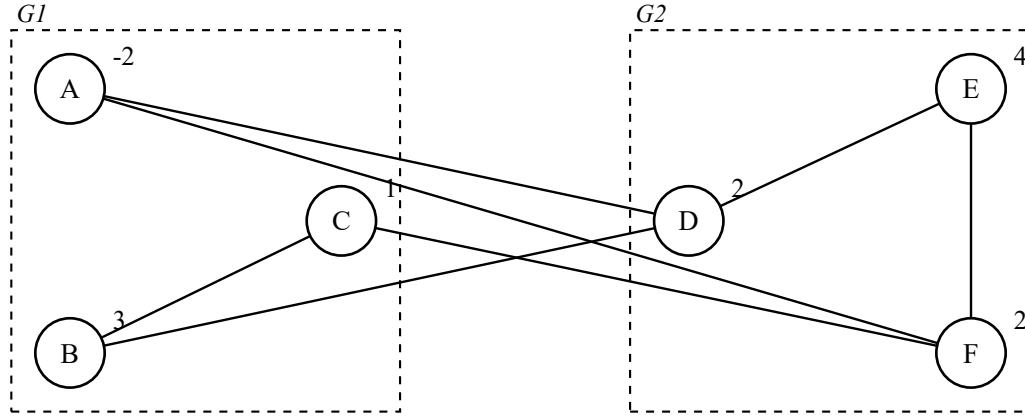


Figure. Graph G

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

6. (GraphSAINT) Consider an undirected graph G of six nodes given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

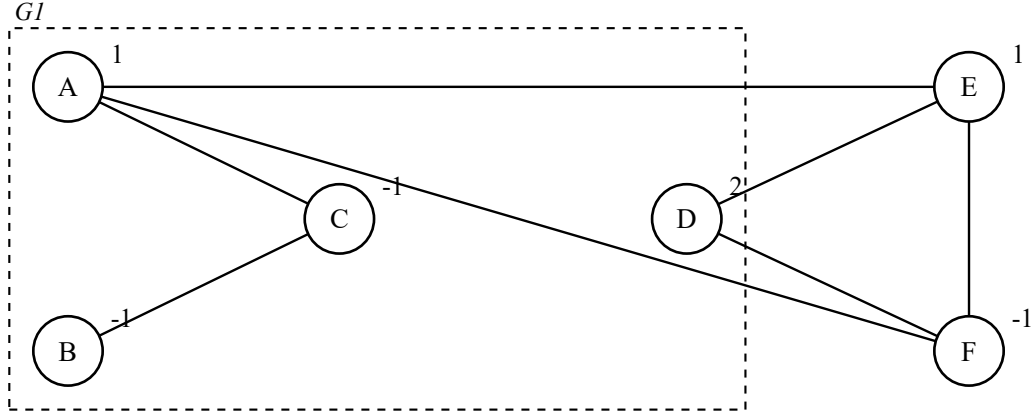


Figure. Graph G

- How does GraphSAINT's sampling differ from node-wise or layer-wise sampling?
- Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 4 \\ -3 \\ -3 \\ -3 \\ 3 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} H^{(0)}), \sigma(\tilde{A} H^{(1)}), \dots, \sigma(\tilde{A} H^{(k-1)}))$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem.
- Calculate $\tilde{A}$ and $H^{(1)}$.
- Calculate the output representations at layer $k = 2$.

8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -3 \\ 3 \\ 3 \\ 0 \\ -1 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $\tilde{A} = D^{-1}A$, I_n is the identity matrix, $\alpha = 0.2$, $\beta = 0.4$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- What are the two major components introduced in GCNII to overcome the depth limitation?
- Calculate $\tilde{A}$.
- Calculate the output representations at layer $k = 1$.

9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -3 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection at layer (k) is:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain why a residual connection allows a GCN to be made deeper.
- Calculate the output representations at layer $k = 1$ and $k = 2$.
- Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.

Practice Set 4

10. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

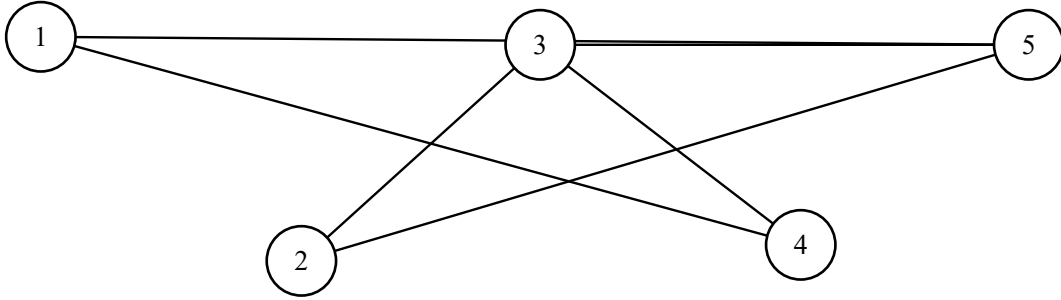


Figure. Graph G

- Explain the key assumption behind relational classification, and state one limitation.
- Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- Calculate $P(Y_v)$ after the 2nd iteration, using the previous iteration's values only.

11. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 3 \\ 3 \\ 2 \\ 0 \\ 2 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$ and the t_l nodes are drawn once per layer from $q(u) \propto \|\tilde{A}(:, u)\|^2$, with $\text{ReLU}(x) = \max(0, x)$.

- How does layer-wise sampling differ from the node-wise sampling of GraphSAGE?
- Calculate the importance distribution $q(u)$ over the five nodes.
- Suppose the sampled set is $\{D, E\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

12. (ASGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 0 \\ -1 \\ 2 \\ 4 \\ 2 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $\tilde{A} = D^{-1}A$, $N(v_i) = \sum_j \tilde{A}_{ij}$, $p(u_j | v_i) = \tilde{A}_{ij} / N(v_i)$, $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- What does the proposal distribution depend on in ASGCN that it does not in FastGCN?
- Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = B$.
- Calculate $E_{p(u_j | B)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(B)$.

13. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

APPNP propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.2$. No nonlinearity is applied between steps.

- Explain the teleport mechanism and why it prevents over-smoothing.
- Calculate $\tilde{A}$ and $Z^{(1)}$.
- Calculate $Z^{(2)}$.

14. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 3 \\ -2 \\ 3 \end{bmatrix}$$

GPR-GNN separates feature extraction from topological propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, $K = 2$, $\gamma_0 = 0.5$, $\gamma_1 = 0.4$, $\gamma_2 = -0.2$.

- What can GPR-GNN do that APPNP cannot, and why do negative γ_k matter?
- Calculate $H^{(1)}$ and $H^{(2)}$.
- Calculate Z .

15. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -2 \\ -2 \\ -2 \\ 3 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \parallel_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- State Theorem 1 of MixHop about the two-hop Delta Operator.
- Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- Calculate $H^{(1)}$ and the Delta Operator $\sigma(\hat{A}H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

16. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} -1 & 0 \\ 0 & 0 \\ 4 & 0 \\ -2 & 0 \end{bmatrix}$$

PairNorm normalizes them with a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_i x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_i \|x_i^c\|_2^2} \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- Explain node-wise and feature-wise over-smoothing, and what PairNorm keeps constant.
- Calculate the centered features.
- Calculate the scaling denominator and the final x_i .

17. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 3 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every epoch:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- Which two problems does DropEdge target, and how does it differ from Dropout?
- Calculate the expected number of surviving edges, and $H^{(1)}$ on the full graph.
- Suppose the edges (A, D) and (A, C) are dropped. Recalculate $H^{(1)}$.

1. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

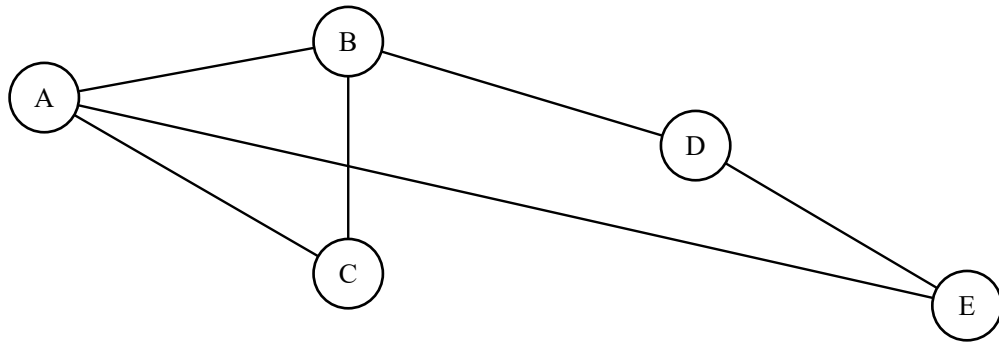


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where $\alpha = 1/p$ if $d(A, x) = 0$, $\alpha = 1$ if $d(A, x) = 1$, and $\alpha = 1/q$ if $d(A, x) = 2$.

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate π_{Bx} .
- Calculate the normalized transition probabilities.

2. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = -0.5, u'_B = -1, u'_C = 0.5, u'_D = 1.$$

The second-order proximity is $p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x)$. Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|B)$, $p_2(B|B)$, $p_2(C|B)$ and $p_2(D|B)$.

3. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ -3 \\ -2 \\ 2 \\ -2 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- a) Explain how information is propagated in a GCN layer.
- b) Calculate $\tilde{A}$.
- c) Calculate the output of the GCN model at layer $k = 1$.

4. (GraphSAGE) Consider an undirected graph G of five nodes given in the following figure. Each node has initial features that are the numbers standing next to it. According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

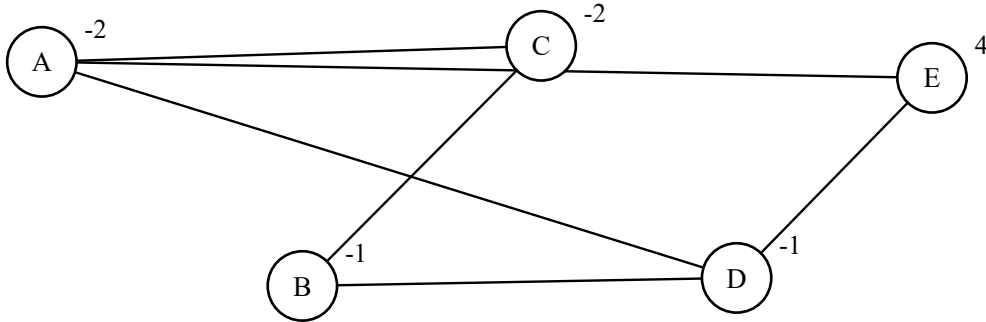


Figure. Graph G

- What is the main idea behind GraphSAGE, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

5. (ClusterGCN) Consider an undirected graph G of six nodes given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

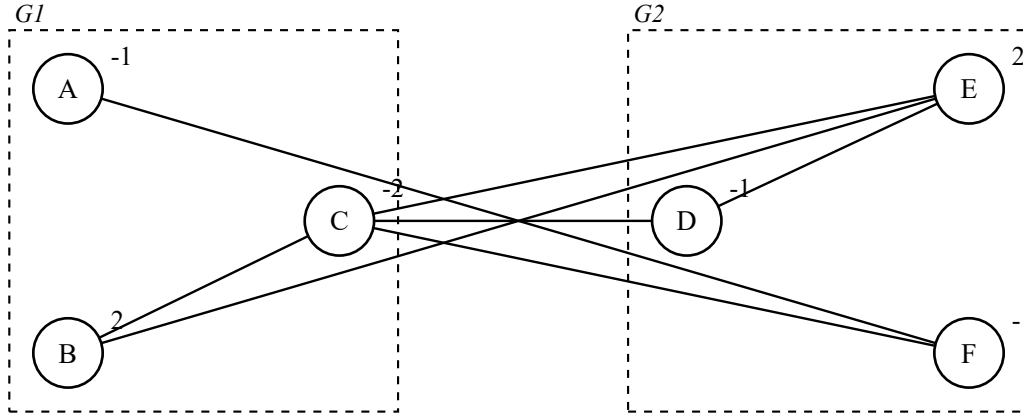


Figure. Graph G

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

6. (GraphSAINT) Consider an undirected graph G of six nodes given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

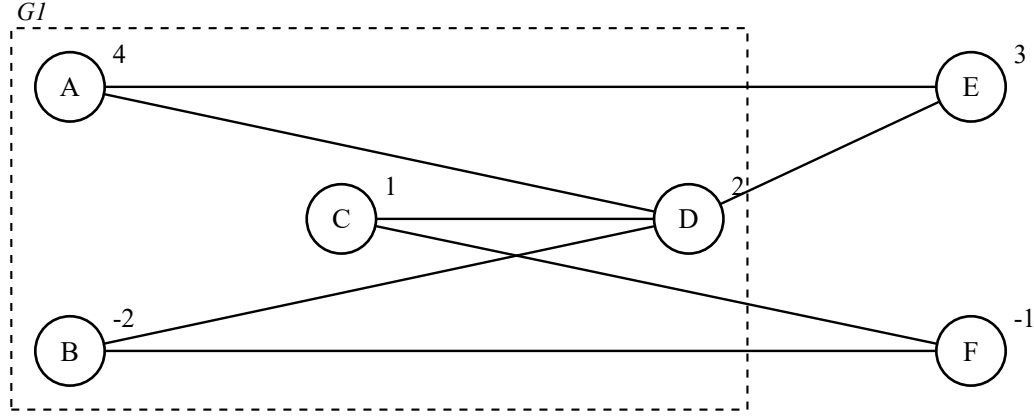


Figure. Graph G

- How does GraphSAINT's sampling differ from node-wise or layer-wise sampling?
- Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ -2 \\ 3 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} H^{(0)}), \sigma(\tilde{A} H^{(1)}), \dots, \sigma(\tilde{A} H^{(k-1)}))$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem.
- Calculate $\tilde{A}$ and $H^{(1)}$.
- Calculate the output representations at layer $k = 2$.

8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ 4 \\ -2 \\ 3 \\ -1 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $\tilde{A} = D^{-1}A$, I_n is the identity matrix, $\alpha = 0.5$, $\beta = 0.5$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- What are the two major components introduced in GCNII to overcome the depth limitation?
- Calculate $\tilde{A}$.
- Calculate the output representations at layer $k = 1$.

9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 4 \\ -2 \\ 0 \\ -2 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection at layer (k) is:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain why a residual connection allows a GCN to be made deeper.
- Calculate the output representations at layer $k = 1$ and $k = 2$.
- Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.

Practice Set 5

10. (Relational Classification) Consider an undirected graph G of five nodes given in the following figure. Nodes 1 and 5 are labelled ($P(Y_1) = 1, P(Y_5) = 0$) and the others are unlabelled and initialized to 0.5. The relational classifier updates every unlabelled node as:

$$P(Y_v = c) = (\sum_{(v,u) \in E} A_{v,u} P(Y_u = c)) / (\sum_{(v,u) \in E} A_{v,u})$$

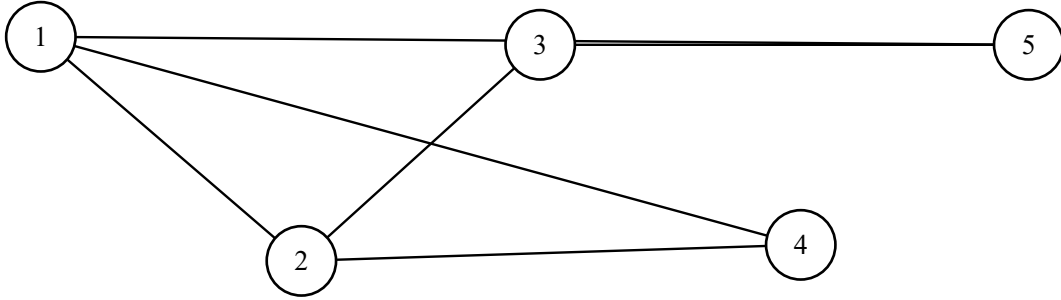


Figure. Graph G

- Explain the key assumption behind relational classification, and state one limitation.
- Calculate $P(Y_v)$ for every unlabelled node after the 1st iteration.
- Calculate $P(Y_v)$ after the 2nd iteration, using the previous iteration's values only.

11. (FastGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. FastGCN replaces the full neighbourhood sum of a GCN layer by a layer-wise Monte-Carlo estimate:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} -2 \\ 4 \\ 4 \\ -1 \\ -2 \end{bmatrix}$$

$$h^{(l+1)}(v) = \sigma \left(\frac{1}{t_l} \sum_{j=1}^{t_l} \tilde{A}(v, u_j) h^{(l)}(u_j) \right)$$

where $\tilde{A} = D^{-1}A$ and the t_l nodes are drawn once per layer from $q(u) \propto \|\tilde{A}(:, u)\|^2$, with $\text{ReLU}(x) = \max(0, x)$.

- How does layer-wise sampling differ from the node-wise sampling of GraphSAGE?
- Calculate the importance distribution $q(u)$ over the five nodes.
- Suppose the sampled set is $\{B, E\}$ ($t_l = 2$). Calculate the estimate of $h^{(1)}(A)$.

12. (ASGCN) Consider the undirected graph G with adjacency matrix A and node feature vector $h^{(0)}$ below. ASGCN rewrites a GCN layer as an expectation:

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$h^{(0)} = \begin{bmatrix} 4 \\ 4 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

$$h^{(l+1)}(v_i) = \sigma(W^{(l)} N(v_i) E_{p(u_j | v_i)}[h^{(l)}(u_j)])$$

where $\tilde{A} = D^{-1}A$, $N(v_i) = \sum_j \tilde{A}_{ij}$, $p(u_j | v_i) = \tilde{A}_{ij} / N(v_i)$, $W^{(l)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- What does the proposal distribution depend on in ASGCN that it does not in FastGCN?
- Calculate $N(v_i)$ and $p(u_j | v_i)$ for the parent node $v_i = E$.
- Calculate $E_{p(u_j | E)}[h^{(0)}(u_j)]$ and the resulting $h^{(1)}(E)$.

13. (APPNP) Given a graph with an adjacency matrix A and a prediction matrix H produced by an MLP as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

APPNP propagates the predictions with a teleport probability α :

$$Z^{(0)} = H, \quad Z^{(k+1)} = (1 - \alpha) \tilde{A} Z^{(k)} + \alpha H$$

where $\tilde{A} = D^{-1}A$ and $\alpha = 0.2$. No nonlinearity is applied between steps.

- Explain the teleport mechanism and why it prevents over-smoothing.
- Calculate $\tilde{A}$ and $Z^{(1)}$.
- Calculate $Z^{(2)}$.

14. (GPR-GNN) Given a graph with an adjacency matrix A and hidden features $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -1 \\ 1 \\ -2 \\ 2 \end{bmatrix}$$

GPR-GNN separates feature extraction from topological propagation:

$$H^{(k)} = \tilde{A} H^{(k-1)}, \quad Z = \sum_{k=0}^K \gamma_k H^{(k)}$$

where $\tilde{A} = D^{-1}A$, $K = 2$, $\gamma_0 = 0.7$, $\gamma_1 = 0.2$, $\gamma_2 = -0.3$.

- What can GPR-GNN do that APPNP cannot, and why do negative γ_k matter?
- Calculate $H^{(1)}$ and $H^{(2)}$.
- Calculate Z .

15. (MixHop) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -3 \\ 1 \\ 0 \end{bmatrix}$$

A MixHop layer concatenates several powers of the adjacency matrix:

$$H^{(i+1)} = \parallel_{j \in P} \sigma(\hat{A}^j H^{(i)} W_j^{(i)})$$

where $P = \{0, 1, 2\}$, $\hat{A} = D^{-1}A$, $W_j^{(i)} = 1$ and $\text{ReLU}(x) = \max(0, x)$.

- State Theorem 1 of MixHop about the two-hop Delta Operator.
- Calculate $\hat{A}^0 H^{(0)}$, $\hat{A}^1 H^{(0)}$ and $\hat{A}^2 H^{(0)}$.
- Calculate $H^{(1)}$ and the Delta Operator $\sigma(\hat{A}H^{(0)}) - \sigma(\hat{A}^2 H^{(0)})$.

16. (PairNorm) After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} -2 & 4 \\ 1 & -1 \\ -1 & 1 \\ -1 & 3 \end{bmatrix}$$

PairNorm normalizes them with a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_i x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_i \|x_i^c\|_2^2} \quad (\text{Scale})$$

with $s = 1$ and $n = 4$.

- Explain node-wise and feature-wise over-smoothing, and what PairNorm keeps constant.
- Calculate the centered features.
- Calculate the scaling denominator and the final x_i .

17. (DropEdge) Consider the complete graph K_4 on nodes A, B, C and D with the adjacency matrix A and initial node feature matrix $H^{(0)}$ below.

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -2 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

DropEdge keeps each undirected edge independently with probability $1 - p$ at every epoch:

$$H^{(1)} = \sigma((D^{(t)})^{-1} A^{(t)} H^{(0)})$$

where $A^{(t)}$ and $D^{(t)}$ are rebuilt from the surviving edges only, $p = 1/3$, and $\text{ReLU}(x) = \max(0, x)$.

- Which two problems does DropEdge target, and how does it differ from Dropout?
- Calculate the expected number of surviving edges, and $H^{(1)}$ on the full graph.
- Suppose the edges (C, D) and (A, D) are dropped. Recalculate $H^{(1)}$.

1. (Node2vec) Consider an undirected graph G given in the following figure. A random walk has just moved from node A to node B , with $p = 2$ and $q = 0.5$.

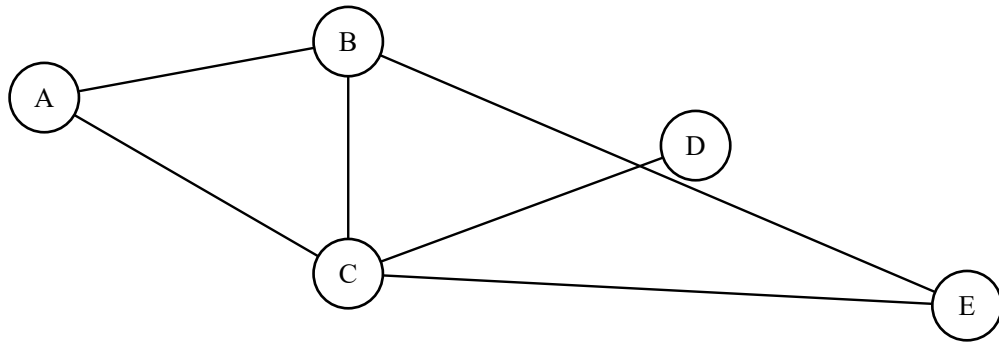


Figure 1. Graph G

The unnormalized transition probability from B to a next node x is $\pi_{Bx} = \alpha_{pq}(A, x)$, where $\alpha = 1/p$ if $d(A, x) = 0$, $\alpha = 1$ if $d(A, x) = 1$, and $\alpha = 1/q$ if $d(A, x) = 2$.

- Explain BFS and DFS, and explain which one a small q makes the walk behave like.
- List all possible next nodes from node B and calculate π_{Bx} .
- Calculate the normalized transition probabilities.

2. (LINE) Consider an undirected graph G with node embeddings

$$u_A = 1, u_B = 0.5, u_C = -0.5, u_D = -1$$

and context embeddings

$$u'_A = 0.5, u'_B = 1, u'_C = -1, u'_D = -0.5.$$

The second-order proximity is $p_2(j|x) = \exp(u'_j u_x) / \sum_k \exp(u'_k u_x)$. Given $e^{0.25} = 1.28$, $e^{0.5} = 1.65$, $e^{-0.25} = 0.78$, $e^{-0.5} = 0.61$.

- a) What is second-order proximity, and how does it differ from first-order proximity?
- b) Calculate $p_2(A|B)$, $p_2(B|B)$, $p_2(C|B)$ and $p_2(D|B)$.

3. (GCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 0 \\ 4 \\ 0 \\ -1 \\ -3 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how information is propagated in a GCN layer.
- Calculate $\tilde{A}$.
- Calculate the output of the GCN model at layer $k = 1$.

4. (GraphSAGE) Consider an undirected graph G of five nodes given in the following figure. Each node has initial features that are the numbers standing next to it. According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node i at layer k can be updated as:

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$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

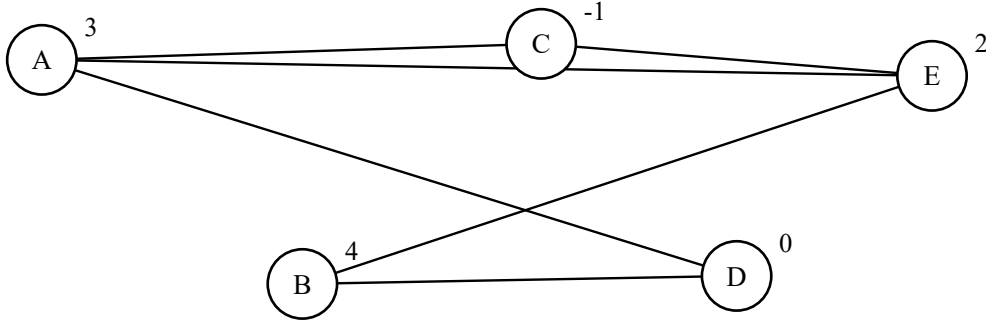


Figure. Graph G

- What is the main idea behind GraphSAGE, and how does it differ from traditional GCNs?
- Calculate the feature of each node at $k = 1$.
- Calculate a graph-level embedding h_G by using a 'Mean' global pooling when $k = 1$.

5. (ClusterGCN) Consider an undirected graph G of six nodes given in the following figure. The graph G contains two clusters G_1 and G_2 . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

where $\parallel$ is a concatenation and $\text{ReLU}(x) = \max(0, x)$.

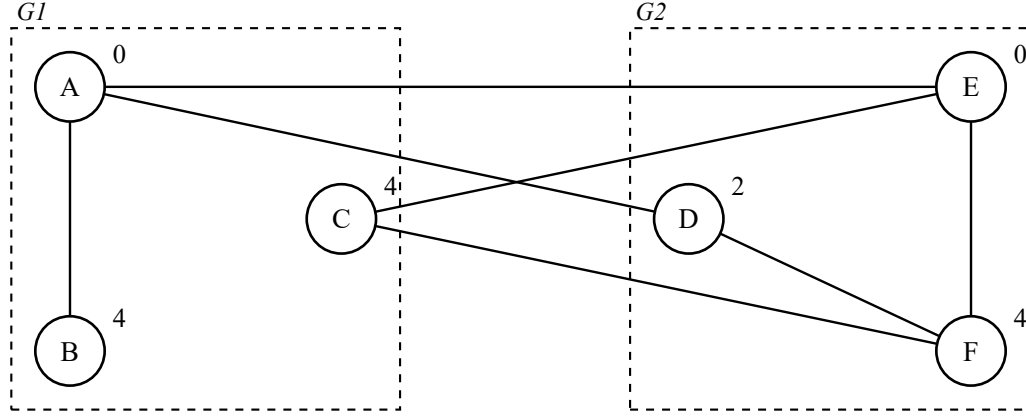


Figure. Graph G

- What is mini-batch training, and why does ClusterGCN need it?
- Calculate the output representations of all nodes at layer $k = 1$.

6. (GraphSAINT) Consider an undirected graph G of six nodes given in the following figure. The graph G has a sampled subgraph G_1 . Each node has initial features that are the numbers standing next to it. According to the GraphSAINT model, the feature of a node i at layer k can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}(h_i^{(k-1)} \parallel h_{N(i)}^{(k)})$$

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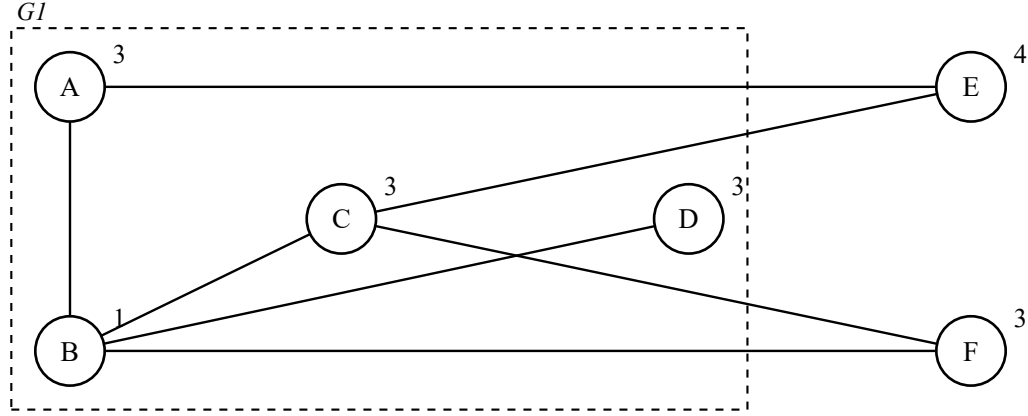


Figure. Graph G

- How does GraphSAINT's sampling differ from node-wise or layer-wise sampling?
- Calculate the output representations of nodes A, B, C and D at layer $k = 1$.

7. (JK Network) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -1 \\ 4 \\ 2 \\ -1 \end{bmatrix}$$

Assume that the output of a JK network model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \max(\sigma(\tilde{A} H^{(0)}), \sigma(\tilde{A} H^{(1)}), \dots, \sigma(\tilde{A} H^{(k-1)}))$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain how Jumping Knowledge Networks help mitigate the over-smoothing problem.
- Calculate $\tilde{A}$ and $H^{(1)}$.
- Calculate the output representations at layer $k = 2$.

8. (GCNII) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -3 \\ 3 \\ 4 \\ -1 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer (k) can be calculated as:

$$H^{(k)} = \sigma[((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)})]$$

where $\tilde{A} = D^{-1}A$, I_n is the identity matrix, $\alpha = 0.5$, $\beta = 0.4$, and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- What are the two major components introduced in GCNII to overcome the depth limitation?
- Calculate $\tilde{A}$.
- Calculate the output representations at layer $k = 1$.

9. (DeepGCN) Given a graph with an adjacency matrix A and initial node feature matrix $H^{(0)}$ as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} -3 \\ 1 \\ 2 \\ 0 \end{bmatrix}$$

Assume that the output of a DeepGCN model with a residual connection at layer (k) is:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}) + H^{(k-1)}$$

where $\tilde{A} = D^{-1}A$ and σ is a ReLU function $\text{ReLU}(x) = \max(0, x)$.

- Explain why a residual connection allows a GCN to be made deeper.
- Calculate the output representations at layer $k = 1$ and $k = 2$.
- Compare $H^{(1)}$ with the output of a plain GCN without the residual connection.